VAPO: Value-Model-Based RL for Advance

60.4 on AIME 2024

Surpassing DAPO by 10+ points on the same base model

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Experimental Setting

Base model: Qwen2.5-32B

No supervised fine-tuning data used

Why Value Models for Long Chain-of-Thought Reasoning?

Value-Model-Free (GRPO, DAPO)

- Group-level reward signal
- Uniform trajectory-level advantage
- Weak token-level credit assignment
- Limited handling of long CoT heterogeneity

Historical issue:

Value-model-based RL underperformed due to bias and instability

Value-Model-Based (VAPO)

- Token-level value estimation
- Finer-grained credit assignment
- Better temporal reasoning optimization
- Enables bias-variance tuning by sequence leng

VAPO result:

Addresses key failure modes and reaches 60.4

Three Core Challenges in Value-Model-Based RL for Long-CoT

1) Value Bias

Long trajectories + bootstrapping errors cause compounding bias.

Observed issue:
Unreliable value targets for long-horizon reasoning.

2) Heterogeneous Lengths

Response lengths vary from short to very long CoT, requiring different bias-variance tradeoffs.

Single fixed λ in GAE is suboptimal across lengths.

3) Reward Sparsity

Outcome rewards are sparse; longer CoT worsens temporal credit assignment difficulty.

Effect:
Slow, unstable policy improvement without shaping.

VAPO Framework: Techniques Mapped to Core Challenges

Challenges

C1 Value bias

C2 Length heterogeneity

C3 Reward sparsity

VAPO Components

- Value-Pretraining
- Decoupled-GAE
- Length-adaptive GAE
- Clip-Higher
- Token-level Policy Gradient Loss
- Positive Example LM Loss
- Group-Sampling

Primary Effect

Value-Pretraining → C1

Decoupled-GAE → C1,C3

Length-adaptive GAE → C2

Clip-Higher → C3

Token-level loss → C3

Positive LM loss → C1,C3

Group-Sampling → C3

Length-Adaptive GAE: Dynamic λ for Variable Response Lengths

Core idea: adapt GAE λ to response length instead of using a single fixed λ.

Short responses → lower λ → lower variance.

Long responses → higher λ → lower bias for long-horizon credit assignment.

Result: better bias-variance tradeoff across heterogeneous sequence lengths.

Short CoT Regime

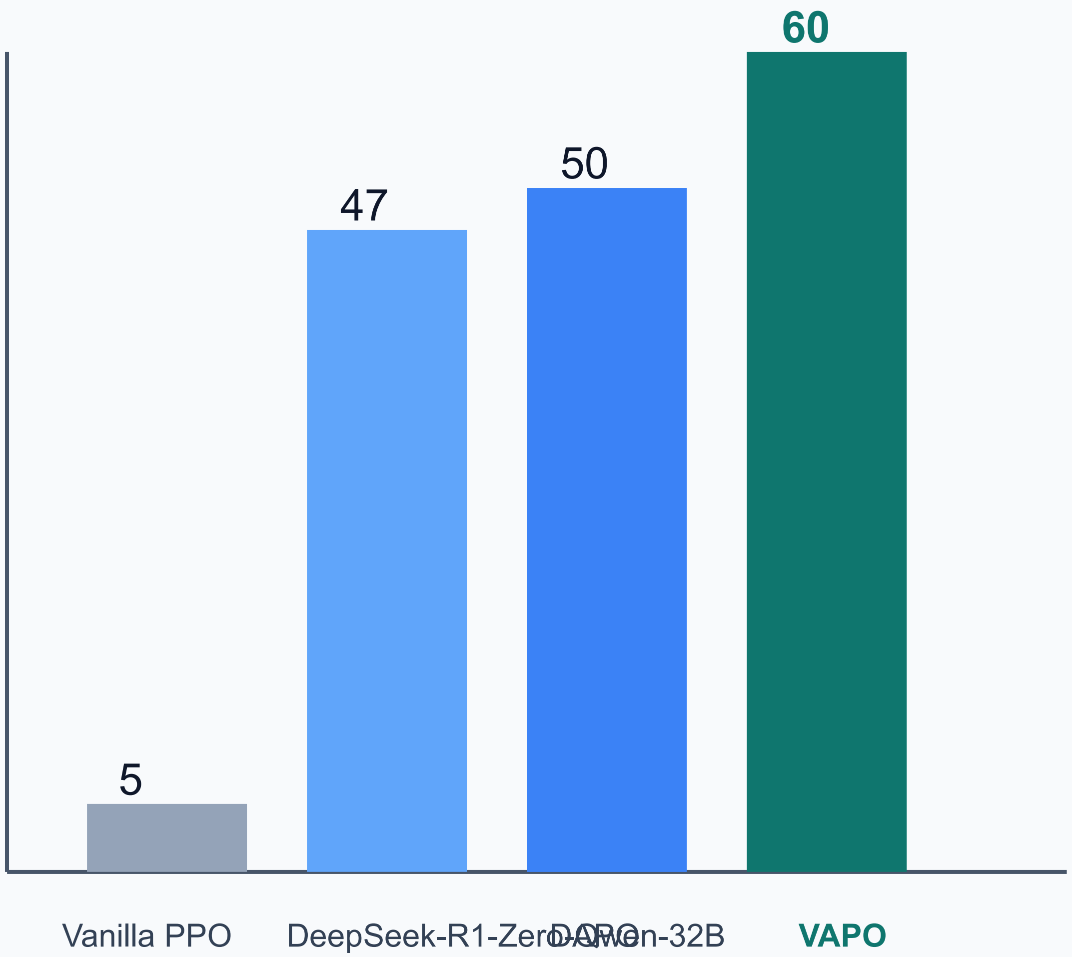
- Lower λ setting
- Stable updates with reduced variance
- Prevents noisy value targets

Long CoT Regime

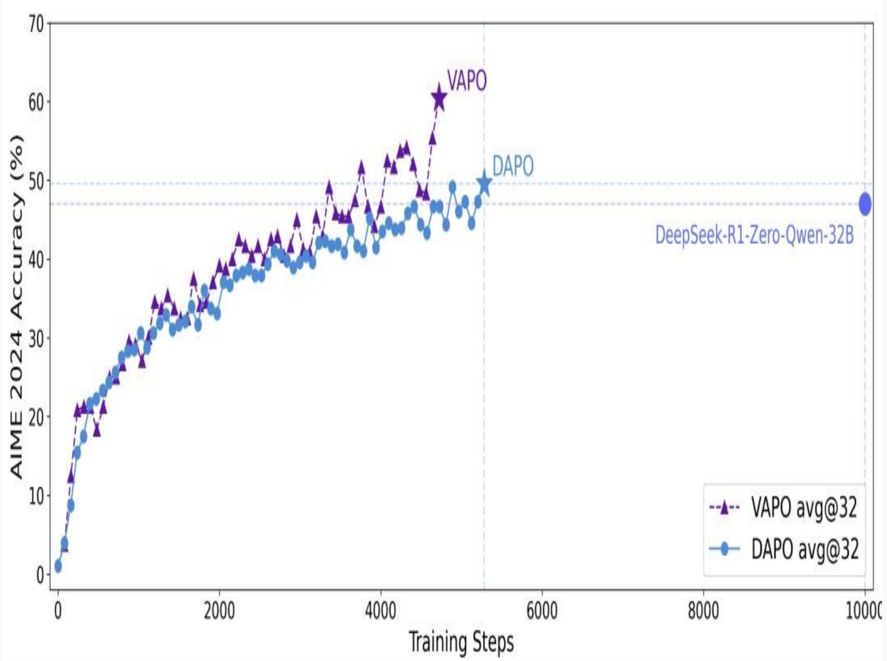
- Higher λ setting
- Less bootstrapping bias
- Better long-horizon advantage estimates

Main Result: VAPO Scores 60 on AIME 2024

10+ points above DAPO and DeepSeek-R1-Zero-Qwen-32B



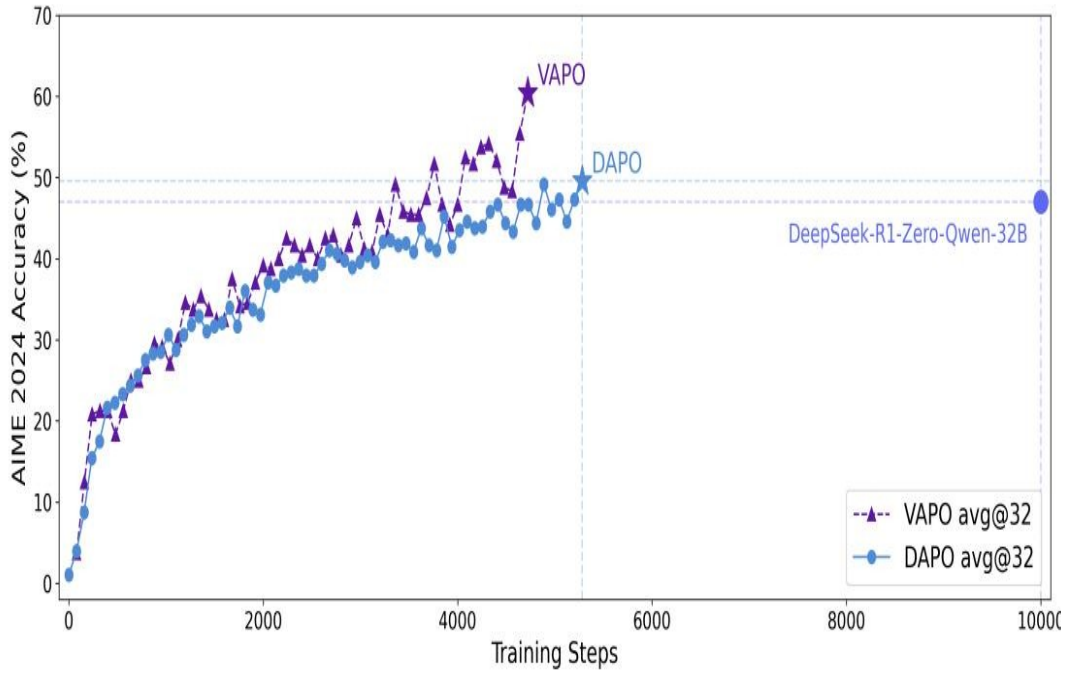
Training Curve (Figure 2)



VAPO peaks at 60.4 around step 5,000.
DAPO peaks near 50 around step 5,500.

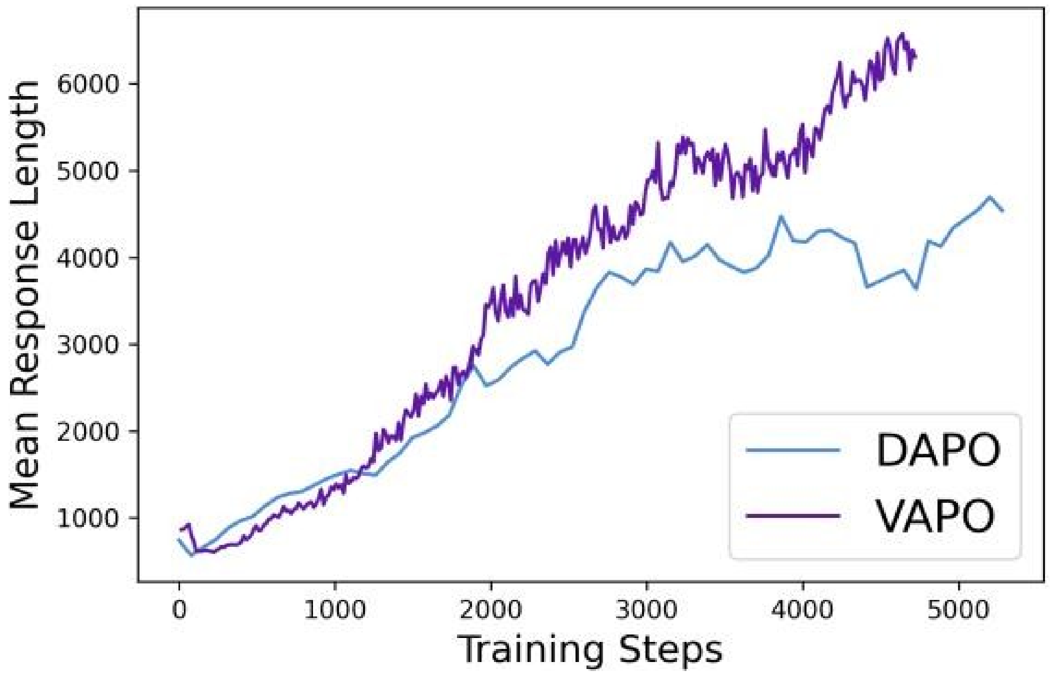
Training Efficiency and Stability

AIME Accuracy vs Steps (Figure 2)



SOTA reached in ~5,000 steps (VAPO).
No training crashes across multiple runs.

Mean Response Length (Figure 3)



VAPO reaches ~6,000 tokens vs DAPO ~4,500.
Suggests richer learned reasoning chains.

Ablation: Component Contributions to AIME 2024

Value-Pretraining is the dominant contributor (60 → 11 when removed)

Configuration	AIME 2024	
VAPO (full)	**60**	
w/o Value-Pretraining	**11**	(-49)
w/o Decoupled-GAE	**33**	(-27)
w/o Length-adaptive GAE	**45**	(-15)

Interpretation: every component contributes; value pretraining is prerequisite for strong value-model-based RL performance in long-CoT reasoning.

Key Takeaways and Future Implications

1. Value-model-based RL can beat value-free baselines.

VAPO reaches 60.4 vs DAPO 50 and DeepSeek-R1-Zero-Qwen-32B 47.

2. Length-adaptive GAE is critical for heterogeneous lengths.

Removing it reduces AIME score from 60 to 45.

3. Value-Pretraining is the highest-impact component.

Without it, score collapses from 60 to 11.

4. Efficiency and stability are first-class outcomes.

SOTA in ~5K steps, with zero reported training crashes across runs.

Implication: Reinvest in value models for next-gen reasoning RL.